

Verdan HMS – Group Report

About the project:

Verdan HMS is a management solution that manages multiple hotels and has been developed by us as part of our coursework. The system caters to four types of users – Guest, Staff, Manager and Admin. The features provided include room search, booking of rooms with services option, check-in and check-out of guests, generating invoices, and generation of room occupation report and income report.

The Team

Raish – Group Leader and Head Developer

Vishal – Developer

Shahudhaan – Developer

How we divided the work

We split the system into three vertical workstreams so each person could own a feature area end-to-end — database, server, UI and tests:

- **Dev1** — authentication, security and the app shell was handled by the group leader Raaish. This covered login, registration, password policy, account lockout and the shared navigation layout every other page sits inside.
- **Dev2** — hotels, rooms and the guest booking experience and **Dev3** — staff operations, billing, reports and audit logging were split between Vishal and Shahudhaan, who coordinated closely because their work often touched the same Booking records.

How we worked

This project lasted about three weeks. In the first two weeks, all time was spent on planning — agreeing on the architecture, dividing responsibilities, creating database schematics, splitting ownership, and writing the individual developers' task list files (dev1.md, dev2.md, dev3.md)

that were both our backlog and acceptance criteria. In the last week, everyone just implemented his piece of software end-to-end and merged his work.

We communicated daily via Discord. There were no strict daily standup meetings since they would only happen when someone encountered a blocker or completed a certain task. All changes in the code base were merged after review of another member via pull requests on GitHub. As for tooling, there was none — not Trello, not Linear, not Figma — we had everything we needed in the markdown task files and pull requests discussion.

Challenges we faced

There were several hurdles during our development process. As our UI library considers the default slot of a modal to be the trigger button instead of the modal content itself, initially all our six modals had appeared empty until we rewrote them. The problem of page routing led us to another roadblock since listing pages could exist only in index.vue file for the route to render properly. Another issue related to login session which did not include the hotel ID associated with each user caused dashboard pages to display nothing.

We also underwent two design overhauls and shifted from using cards to tables. Generating test data was yet another hassle; our database had to be re-seeded several times until we got a set of bookings which included past, present and future cases covering all edge cases like peak prices, cancellation fees, double booking checks, and account lock-out.

Outcome

We implemented everything that was required of us, supported by automated unit tests and complete audit logs. Peer reviews of our pull requests and rapid iteration enabled us to identify issues before they became expensive to solve.